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|---|---|---|
| | **GLOBAL ECONOMICS**<br>**Grade: 10 - Learning Evidence 3**<br>**DECISION-MAKING UNDER UNCERTAINTY**<br>**Teacher: Nicolás López Cuéllar** | **1st TERM**<br>**2026–2027** |

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|---|---|
| **Learning objective:**<br>Compare decision criteria and their limitations, and construct and interpret decision trees for staged filtering under uncertainty. | **Assessment criteria:**<br>**C8**: Compares the Maximax, Maximin, Minimax Regret, and EMV/Bayes decision criteria by contrasting their information requirements, risk orientations, recommended alternatives, and limitations.<br>**C9**: Constructs and interprets decision trees for staged filtering rules, applying decision criteria in the specified order to identify the final alternative.<br>**C10**: Evaluates economic and financial decision-making scenarios by selecting and justifying the most appropriate decision criterion according to the information available, risk attitude, and decision context. |

**Name:** _____ **Grade:** _____ **Date:** _____

## Criteria assessment

This exam assesses C8 and C9, each worth 6 points. A **minimum of 4 points per criterion** is required to pass. Show all relevant calculations and justify your answers. Partial credit may be awarded for correct procedures despite arithmetic errors; answers without demonstrated procedures receive no credit. A 1-point penalty per criterion applies if the student's name is missing or the assessment is submitted late. C10 is assessed separately.

| Question | C8 | C9 |
|---|---|---|
| 1 | 3 | – |
| 2 | 3 | – |
| 3 | – | 2 |
| 4 | – | 2 |
| 5 | – | 2 |
| **Total** | **6** | **6** |

## 1. Repair Workshop Plans [C8-3]

A bicycle repair workshop selects its capacity under uncertain demand. Profits are in thousand USD.

| Alternative | High | Normal | Low |
|---|---|---|---|
| $P(S_j)$ | 0.2 | 0.6 | 0.2 |
| Large ($A_1$) | 120 | 25 | -20 |
| Flexible ($A_2$) | 85 | 65 | -10 |
| Basic ($A_3$) | 50 | 45 | 35 |
| Partner ($A_4$) | 70 | 75 | 20 |

Decision statistics (Regret = maximum regret):

| Alt. | Max | Min | Regret | EMV |
|---|---|---|---|---|
| $A_1$ | 120 | -20 | 55 | 35 |
| $A_2$ | 85 | -10 | 45 | 54 |
| $A_3$ | 50 | 35 | 70 | 44 |
| $A_4$ | 75 | 20 | 50 | 63 |

**(a)** Identify each criterion's recommendation and deciding statistic value. **[1]**

**(b)** A subsidy $s$ (integer, $0 \leq s \leq 50$) increases the low-demand payoffs of $A_1$ and $A_2$ by $s$. An alternative is eligible only if all its payoffs are nonnegative. Determine each alternative's eligibility based on the value of $s$ and find the smallest $s$ making $A_2$ the sole Maximin choice. **[2]**

## 2. Community Market Stall [C8-3]

An organizer selects a market stall under uncertain attendance. Profits are in hundred USD.

| Alternative | High | Low |
|---|---|---|
| $P(S_j)$ | 0.4 | 0.6 |
| Large ($C_1$) | 80 | 10 |
| Compact ($C_2$) | 48 | 32 |
| Standard ($C_3$) | 70 | 25 |
| Premium ($C_4$) | 90 | -5 |

Decision statistics (Regret = maximum regret):

| Alternative | Max | Min | Regret | EMV |
|---|---|---|---|---|
| $C_1$ | 80 | 10 | 22 | 38 |
| $C_2$ | 48 | 32 | 42 | 38.4 |
| $C_3$ | 70 | 25 | 20 | 43 |
| $C_4$ | 90 | -5 | 37 | 33 |

**(a)** Identify each criterion's recommendation and deciding statistic value. **[1]**

**(b)** A sponsor replaces $C_2$'s high-attendance payoff of 48 with $g$ (integer, $48 \leq g \leq 80$). An alternative is eligible only if all its payoffs are at least 20. Determine each alternative's eligibility based on $g$ and find the smallest $g$ making $C_2$ the sole Minimax Regret choice among eligible alternatives. **[2]**

## 3. Book Fair Stock [C9-2]

A bookseller selects a stock plan $D_1$–$D_6$. Net profits are in hundred USD. The decision statistics are summarized below:

| Alternative | Max | Min | Regret | EMV |
|---|---|---|---|---|
| $D_1$ | 100 | 0 | 40 | 65 |
| $D_2$ | 90 | 10 | 30 | 65 |
| $D_3$ | 80 | 20 | 20 | 65 |
| $D_4$ | 70 | 30 | 30 | 59 |
| $D_5$ | 60 | 40 | 40 | 53 |
| $D_6$ | 50 | 30 | 50 | 43 |

Apply these stages in order:

1. EMV/Bayes: Best 3.
2. Maximax: Best 2.
3. Maximin: Final selection.

Illustrate each filtering stage, showing survivors and eliminations.

## 4. Refill Shop Expansion [C9-2]

A refill shop selects an expansion plan $F_1$–$F_6$. Net profits are in thousand USD. The decision statistics are summarized below:

| Alternative | Max | Min | Regret | EMV |
|---|---|---|---|---|
| $F_1$ | 120 | 0 | 45 | 61 |
| $F_2$ | 105 | 15 | 30 | 64.5 |
| $F_3$ | 95 | 25 | 25 | 68.5 |
| $F_4$ | 85 | 35 | 35 | 70 |
| $F_5$ | 75 | 45 | 45 | 64 |
| $F_6$ | 65 | 40 | 55 | 55 |

Apply these stages in order:

1. Minimax Regret: Best 5.
2. EMV/Bayes: Best 4.
3. Maximax: Best 2.
4. Maximin: Final selection.

Illustrate each filtering stage, showing survivors and eliminations.

## 5. School Supply Orders [C9-2]

A school supplier selects an order plan $H_1$–$H_6$. Net profits are in hundred USD. The decision statistics are summarized below:

| Alternative | Max | Min | Regret | EMV |
|---|---|---|---|---|
| $H_1$ | 120 | 0 | 45 | 61 |
| $H_2$ | 105 | 15 | 30 | 64.5 |
| $H_3$ | 95 | 25 | 25 | 68.5 |
| $H_4$ | 85 | 35 | 35 | 70 |
| $H_5$ | 75 | 45 | 45 | 64 |
| $H_6$ | 65 | 40 | 55 | 55 |

Apply these stages in order:

1. Maximin: Best 4.
2. Minimax Regret: Best 3.
3. Maximax: Best 2.
4. EMV/Bayes: Final selection.

Illustrate each filtering stage, showing survivors and eliminations.

## C10: Separate activity reminder (unscored)

Complete the individual **Interactive Decision Webpage: C10 Practice** activity and submit the deployed webpage URL. See the separate activity for instructions.